

Semiconductor

1] Conductivity and mobility of semiconductors: -

$$J = (n_e \cdot e \cdot \mu_e + n_h \cdot e \cdot \mu_h) E$$

$$\sigma_{\text{ext}} = (n_e \cdot \mu_e + n_h \cdot \mu_h) e$$

$$\sigma_{\text{int}} = n_i (\mu_e + \mu_h) \cdot e$$

Where,

1. J $\square$ current density
2. n_e and n_h $\square$ density of electrons and holes respectively
3. μ_e and μ_h $\square$ mobility of electrons and holes respectively
4. n_i $\square$ density of electrons/ holes for intrinsic semiconductor
5. σ_{ext} and σ_{int} $\square$ conductivity of electrons and holes respectively

2] Fermi Dirac distribution function: -

$$f(E_c) = \frac{1}{1 + e^{(E_c - E_f)/kT}}$$

where,

E_c = energy of conduction band level

E_f = energy of Fermi level

T = temperature

K = Boltzmann constant

3] Hall effect: -

$$V_H = \frac{B J d}{n_e e}$$

where,

V_H : Hall voltage

B: Magnetic field

J: Current density

$$R_H = \frac{1}{n_e e}$$

where,

R_H : Hall coefficient

$$\mu_e = \sigma R_H$$

where,

σ : Conductivity

μ_e : Mobility in electrons

$$\rho = \frac{1}{\sigma}$$

where,

ρ : resistivity

σ : Conductivity